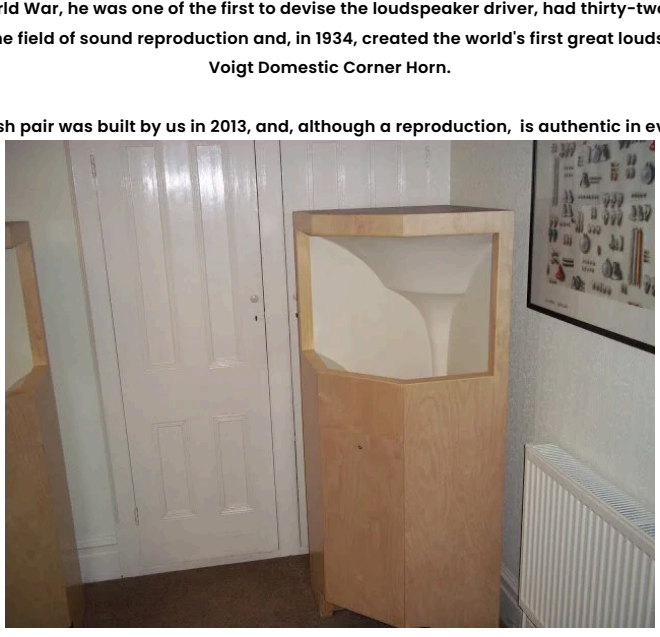
Jul 26 • Written By Iain

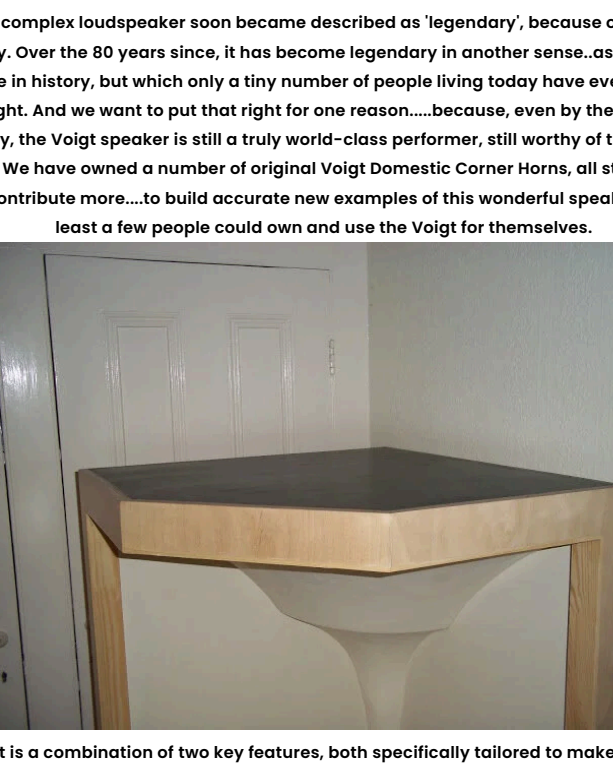
Secrets of the Voigt Speaker - By Peter Skinner

Paul Voigt was one of the true pioneers of audio and loudspeaker technology. Working before the Second World War, he was one of the first to devise the loudspeaker driver, had thirty-two important patents in the field of sound reproduction and, in 1934, created the world's first great loudspeaker: The Voigt Domestic Corner Horn.

This fresh pair was built by us in 2013, and, although a reproduction, is authentic in every detail.

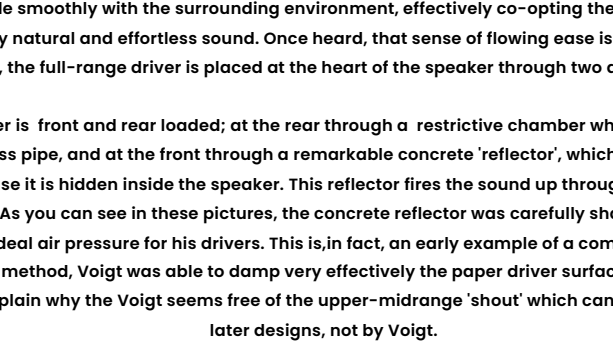


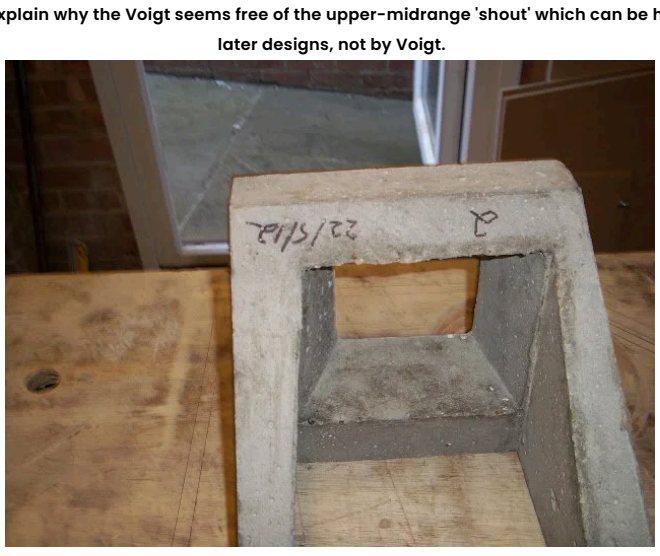
This highly complex loudspeaker soon became described as 'legendary', because of it's exceptional sound quality. Over the 80 years since, it has become legendary in another sense..as a speaker with a massive place in history, but which only a tiny number of people living today have ever heard. We want to put that right. And we want to put that right for one reason.....because, even by the standards of the 21st century, the Voigt speaker is still a truly world-class performer, still worthy of the description 'legendary' . We have owned a number of original Voigt Domestic Corner Horns, all still in use, but we wanted to contribute more....to build accurate new examples of this wonderful speaker. That way ,at least a few people could own and use the Voigt for themselves.

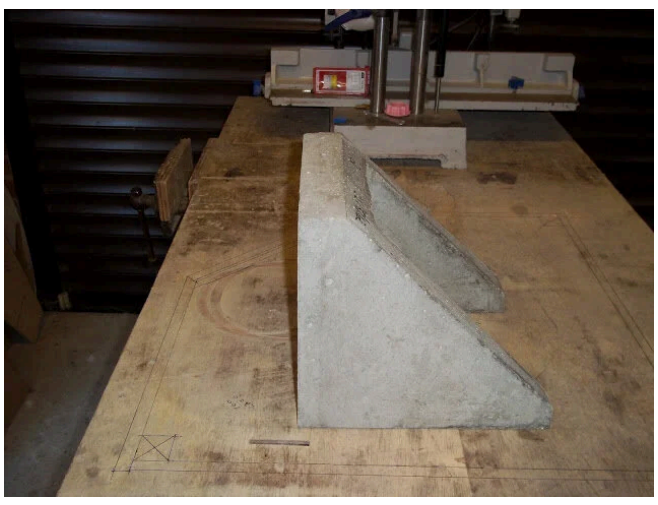


The Voigt is a combination of two key features, both specifically tailored to make the best use of a single full-range driver; this was invented and patented by Voigt, and in time became the family of Lowther drivers. There have been numerous versions of this type of Voigt-driver , but all can be traced back to his original insights.To achieve a fully effective loudspeaker, Voigt used a section of a Tractrix horn (an idea patented by him in1927) together with a quarter-wave, horn-loaded bass section. The horns couple smoothly with the surrounding environment, effectively co-opting the room to offer a remarkably natural and effortless sound. Once heard, that sense of flowing ease is hard to forget. Uniquely, the full-range driver is placed at the heart of the speaker through two access doors.

The driver is front and rear loaded; at the rear through a restrictive chamber which flows into a widening bass pipe, and at the front through a remarkable concrete 'reflector', which has rarely been seen, because it is hidden inside the speaker. This reflector fires the sound up through the section of tratix horn. As you can see in these pictures, the concrete reflector was carefully shaped by Voigt to create the ideal air pressure for his drivers. This is,in fact, an early example of a compression driver. Through this method, Voigt was able to damp very effectively the paper driver surface, front and rear, which may explain why the Voigt seems free of the upper-midrange 'shout' which can be heard in some later designs, not by Voigt.







The question which will naturally occur to people is the obvious one: how can a speaker created in the 1930s (1934), be competitive eighty years later, after decades of technological progress? Well, the UK's most experienced speaker reviewer, Paul Messenger, listened to an early pair of the Voigts recently, and remarked in the magazine 'Stereophile', on the natural and sweet sound and noted: "it makes you wonder whether there has been any real progress". A wide range of highly experienced listeners have been surprised and impressed by what they have experienced,with this speaker, using modern drivers, derived directly from Voigt's original ideas. When I first heard a Voigt, at John Howes' Lowther/Voigt Museum, I was astonished . Later, I spent twelve months using a pair of 1930s Voigts, comparing them with later 'classics' such as Quad electrostatics, Spendor BC1s and even very expensive Wilson Watts and Puppies. Each speaker had its own strengths and weaknesses, but overall the Voigts distinquished themselves. Colleagues felt the same. But there were less than 400 Voigts made, between 1934 and 1954. Many have been ruined by woodworm. Those still around are often a bit fragile, and have become very expensive.

Since production ended in 1954, there have been various attempts to produce a reproduction. These attempts foundered on the absence of any published plans; Voigt insisted, probably rightly, that his design was too complex to be built by amateurs. Instead he offered a greatly simplified 'home constructors' version, which was, in fact, a quite different loudspeaker. There are plenty of these about, but the few attempts at the original design all seemed to have obvious errors.

GET IT RIGHT....OR DON'T BOTHER

All this changed in 2011 when the American horn expert, Bruce Edgar, sent some original Voigt documents to John Howes, at the Voigt/Lowther Museum. A furniture maker of great expertise,Les Bartlett (my brother-in-law) was commissioned by John Howes to make a new pair, using the documents. In fact it turned-out that the documents were essentially a list of parts, and not a plan of how to assemble the speaker. However, Les was also given access to a woodwormy old Voigt, from which he could copy the design. This proved difficult because the Voigt is a complex construction, with numerous angles, curves and unexpected chambers, not to mention the tractrix horn. Les Bartlett spent a great deal of time working-out how it had all been done. By the Summer of 2012, he had built the first new pair of Voigts, and these featured in an article in' 'Hi Fi News'. Subsequently, it was realised that a mistake had been made, and two baffles omitted from the bass cabinet design. This was no-one's fault, the baffles had been missing from the drawings in the original documents, and Les had no way of knowing this. To fix the issue would require forcing open the sealed cabinets and horn, and partially rebuilding the speaker. It was just a bit of bad luck for all concerned, but perhaps not so surprising for a first effort.

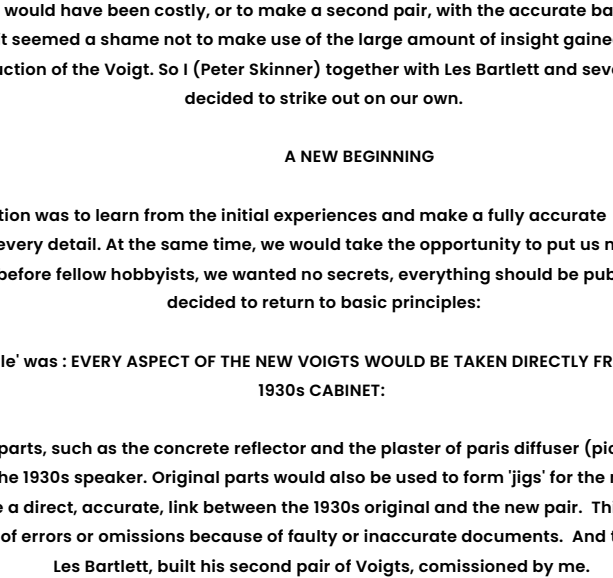
At this point the 'Voigt project' seemed to go quiet, with no obvious plans either to rectify the error, which would have been costly, or to make a second pair, with the accurate bass design. However, it seemed a shame not to make use of the large amount of insight gained into the design and construction of the Voigt. So I (Peter Skinner) together with Les Bartlett and several colleagues, decided to strike out on our own.

A NEW BEGINNING

The intention was to learn from the initial experiences and make a fully accurate pair of Voigts, authentic in every detail. At the same time, we would take the opportunity to put us much information as possible before fellow hobbyists, we wanted no secrets, everything should be publicly shared. We decided to return to basic principles:

The new 'rule' was : EVERY ASPECT OF THE NEW VOIGTS WOULD BE TAKEN DIRECTLY FROM AN ORIGINAL 1930s CABINET:

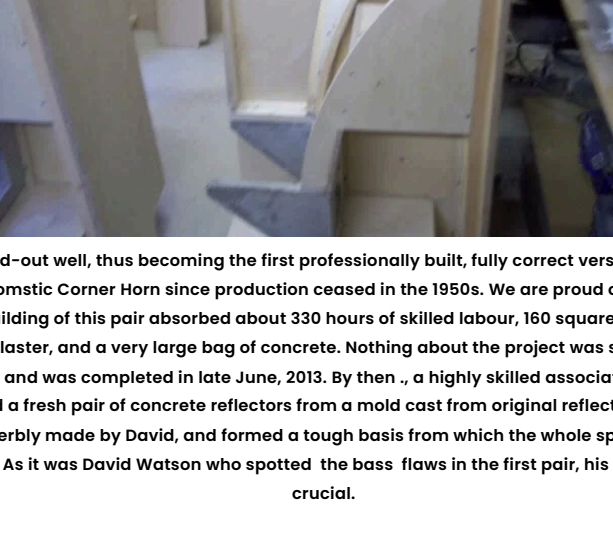
The crucial parts, such as the concrete reflector and the plaster of paris diffuser (pic) would be taken directly from the 1930s speaker. Original parts would also be used to form 'jigs' for the new pair. This way, there would be a direct, accurate, link between the 1930s original and the new pair. This way there would be no chance of errors or omissions because of faulty or inaccurate documents. And this is exactly how Les Bartlett, built his second pair of Voigts, comissioned by me.

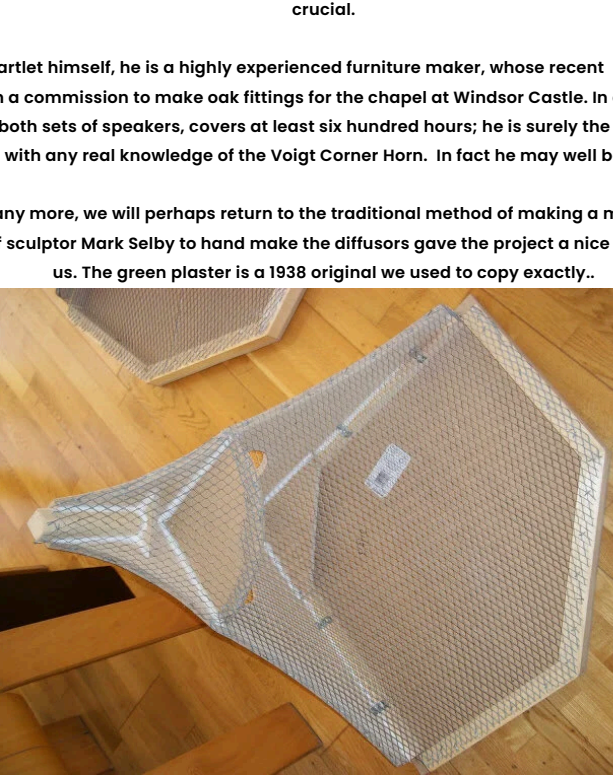


They turned-out well, thus becoming the first professionally built, fully correct version of the Voigt Domstic Corner Horn since production ceased in the 1950s. We are proud of this.
In all, the building of this pair absorbed about 330 hours of skilled labour, 160 square feet of best Birch ply, 50lbs of plaster, and a very large bag of concrete. Nothing about the project was simple. It began in October 2012, and was completed in late June, 2013. By then ,, a highly skilled associate, David Watson, had achieved a fresh pair of concrete reflectors from a mold cast from original reflectors. The new pair were superbly made by David, and formed a tough basis from which the whole speaker can be constructed. As it was David Watson who spotted the bass flaws in the first pair, his contribution was crucial.

As for Les Bartlet himself, he is a highly experienced furniture maker, whose recent projects include involvement in a commission to make oak fittings for the chapel at Windsor Castle. In all, his work on the Voigt project, both sets of speakers, covers at least six hundred hours; he is surely the most experienced craftsman with any real knowledge of the Voigt Corner Horn. In fact he may well be the only one.

If we make any more, we will perhaps return to the traditional method of making a mold. But using the artistic skill of sculptor Mark Selby to hand make the diffusors gave the project a nice touch, at least for us. The green plaster is a 1938 original we used to copy exactly..





Our longer term aim is to prepare full plans, so that expert amateur enthusiasts can make their own Voigts, or pay others to make them. This will take more time to prepare, as it will probably involve commisioning a professional draughtsman. Which won't be cheap for us. When the plans are completed, they will be offered at no charge.

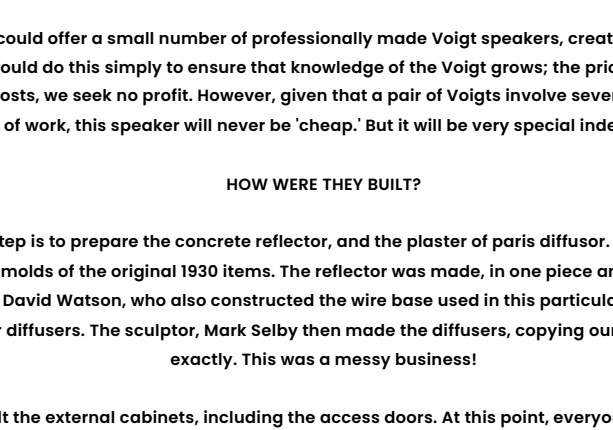
DO YOU WANT A PAIR OF VOIGT DOMESTIC CORNER HORNS

For now we could offer a small number of professionally made Voigt speakers, created to order by Les Bartlett. We would do this simply to ensure that knowledge of the Voigt grows; the price would be aimed to cover our costs, we seek no profit. However, given that a pair of Voigts involve several hundred hours of work, this speaker will never be 'cheap.' But it will be very special indeed.

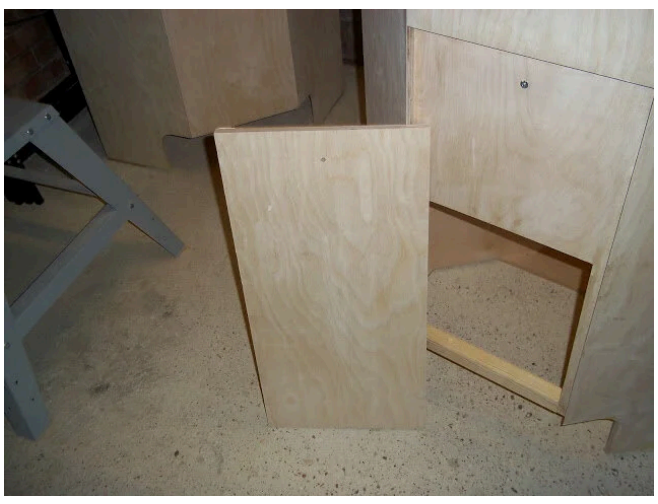
HOW WERE THEY BUILT?

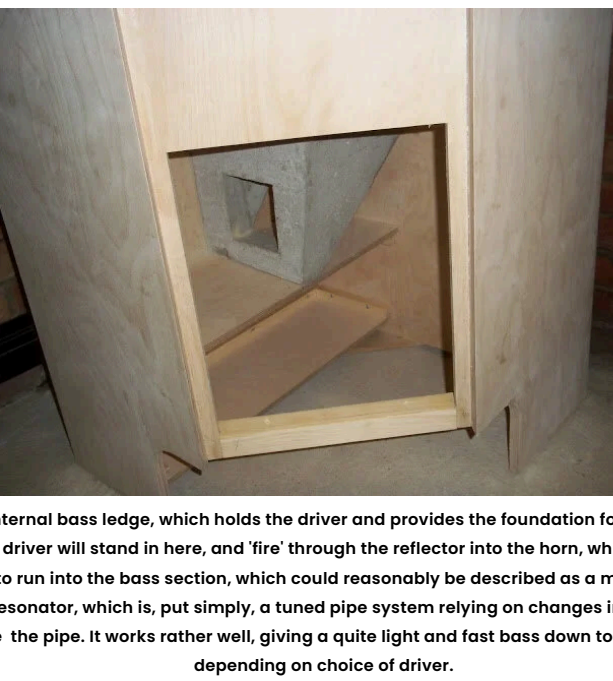
The first step is to prepare the concrete reflector, and the plaster of paris diffusor. Both were taken direct from molds of the original 1930 items. The reflector was made, in one piece and with extreme precision, by David Watson, who also constructed the wire base used in this particular construction of the plaster diffusers. The sculptor, Mark Selby then made the diffusers, copying our 1930s diffusor exactly. This was a messy business!

Next, Les built the external cabinets, including the access doors. At this point, everyone felt pleased at the speed of progress, not realizing that 90% of the work was still ahead.



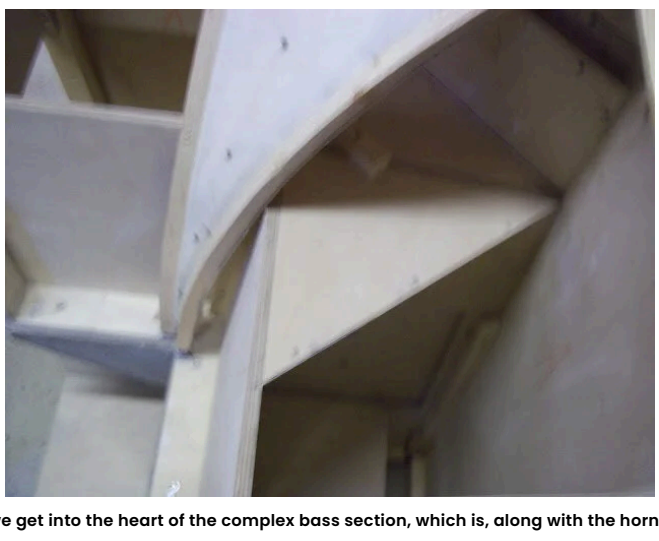
Then the internal bass ledge, which holds the driver and provides the foundation for the initial bass chamber. The driver will stand in here, and 'fire' through the reflector into the horn, while the bass sound waves begin to run into the bass section, which could reasonably be described as a modified version of a hemholtz resonator, which is, put simply, a tuned pipe system relying on changes in the air pressure inside the the pipe. It works rather well, giving a quite light and fast bass down to around 40hz, depending on choice of driver.

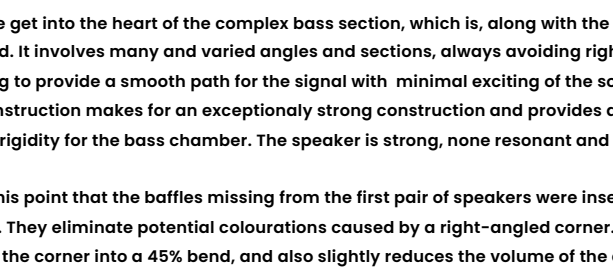




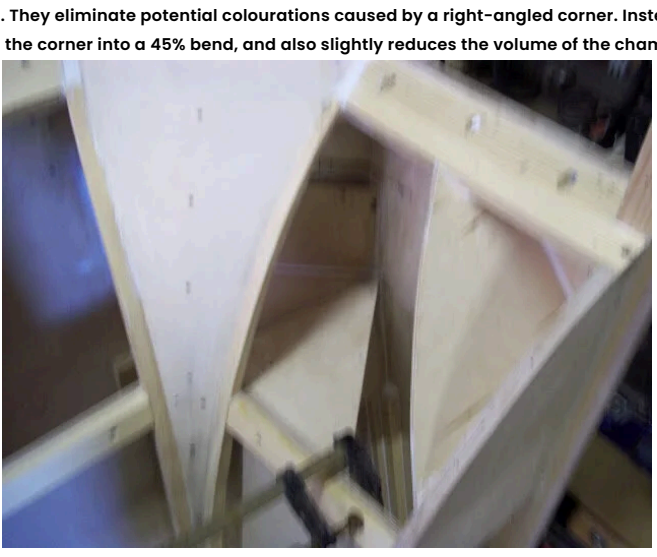
After this, we get into the heart of the complex bass section, which is, along with the horn, the hardest section to build. It involves many and varied angles and sections, always avoiding right-angled corners, the aim being to provide a smooth path for the signal with minimal exciting of the sound wave. As an aside, this construction makes for an exceptionally strong construction and provides a very high degree of rigidity for the bass chamber. The speaker is strong, none resonant and tough.

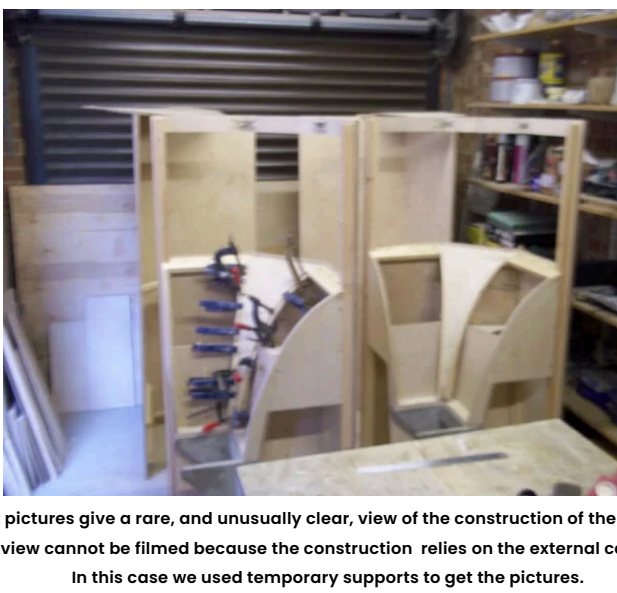
It was at this point that the baffles missing from the first pair of speakers were inserted into the new bass cabinets. They eliminate potential colourations caused by a right-angled corner. Instead the baffle turns the corner into a 45% bend, and also slightly reduces the volume of the chamber.

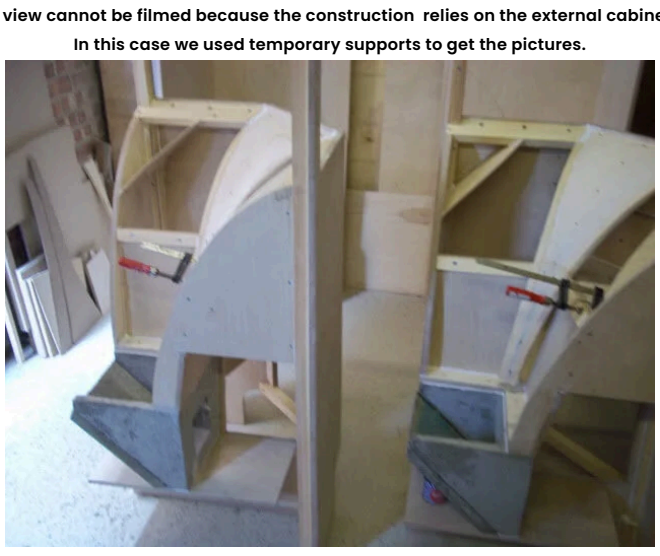


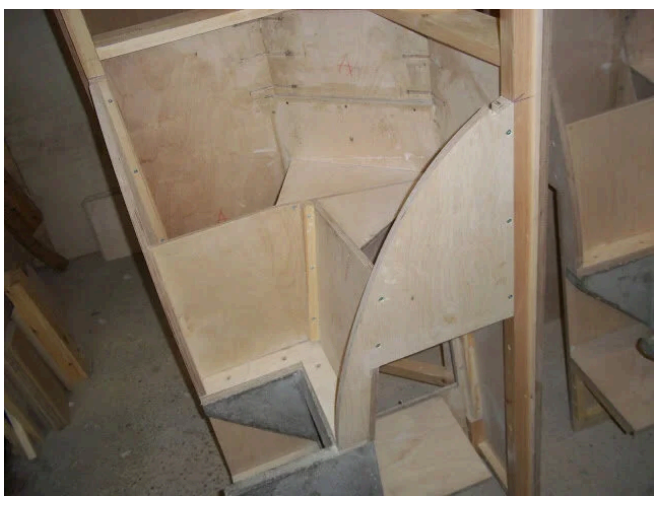


The next pictures give a rare, and unusually clear, view of the construction of the bass chamber. Normally this view cannot be filmed because the construction relies on the external cabinet for support. In this case we used temporary supports to get the pictures.









Now, an interesting bit, where les Bartlett begins to fix the horn. This is far trickier than it might look; the nature of the curves, and a complex frame at the top (which has varied thickness along its length) makes precision vital. In this case, Les built a mock-up to ensure everything fitted. This is the last point at which the construction of the bass chamber can be viewed. After this, it is permanently sealed.

The end approached and life got a bit easier for Les; there were more fittings round the base of the horn, and other little 'fiddly bits', of which there are a lot in the Voigt Domestic Corner Horn.
Then the external cabinet is fitted....made at the start of construction it ends up as one of the last bits to be completed.

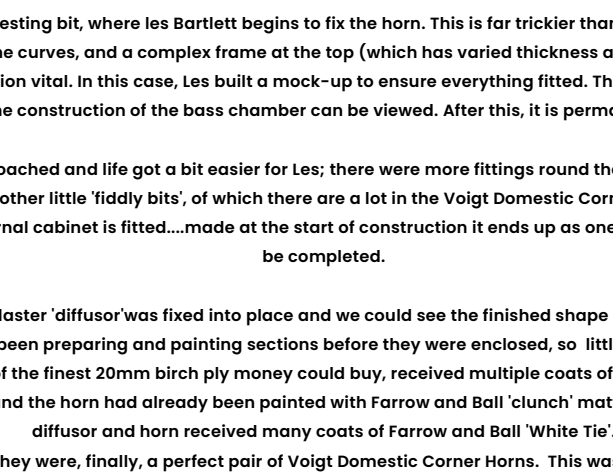
The giant plaster 'diffusor'was fixed into place and we could see the finished shape of the speakers, at last! Les had been preparing and painting sections before they were enclosed, so little remained to do. The cabinet, of the finest 20mm birch ply money could buy, received multiple coats of varnish, the inner sections around the horn had already been painted with Farrow and Ball 'clunch' matt shade, while the diffusor and horn received many coats of Farrow and Ball 'White Tie'.
And there they were, finally, a perfect pair of Voigt Domestic Corner Horns. This was a nice moment.

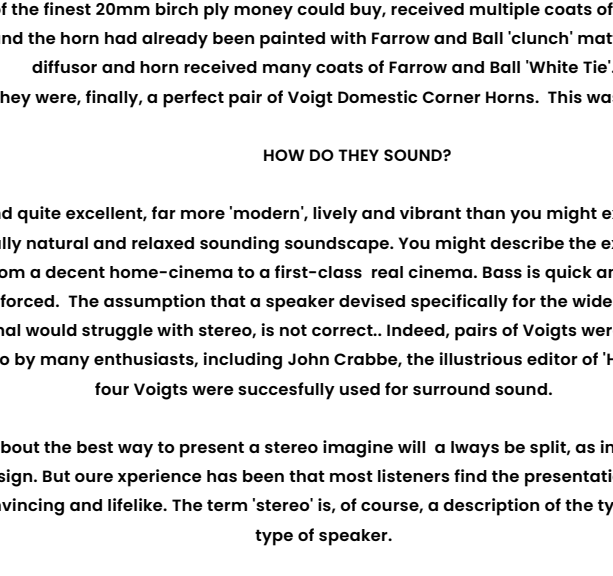
HOW DO THEY SOUND?

They sound quite excellent, far more 'modern', lively and vibrant than you might expect. There is a huge, unusually natural and relaxed sounding soundscape. You might describe the experience as a bit like moving from a decent home-cinema to a first-class real cinema. Bass is quick and fluid, the sound sweet and unforced. The assumption that a speaker devised specifically for the wide-scale dispersion of a mono signal would struggle with stereo, is not correct.. Indeed, pairs of Voigts were used in the early years of stereo by many enthusiasts, including John Crabbe, the illustrious editor of 'Hi Fi News'. Sets of four Voigts were succesfully used for surround sound.

Opinions about the best way to present a stereo imagine will always be split, as in every aspect of speaker design. But oure xperience has been that most listeners find the presentation of the Voigts unusually convincing and lifelike. The term 'stereo' is, of course, a description of the type of signal, not a type of speaker.

Was it worth it? Absolutely. A truly great loudspeaker from one of the original masters. We just hope more people get a chance to hear them.





All pictures and text provided by Peter Skinner. This project is absolutely amazing and a fine tribute to Voigt. I will be eagerly awaiting the Voigt plans. Thank you so much for remaining dedicated to this project and for debuting this information here first.
Thank you Peter Skinner, Les Bartlett, David Watson, and John Howes.

Corner horn • Peter Skinner • Voigt Domestic Corner Horn • Voigt

Iain

Comments (45)

Newest First

Preview Post Comment…

Tamas Abonyi 4 years ago · 0 Likes

Dear Peter Skinner

Please send me the drawings sent by Bruce Edgar if possible. In the Domestic Corner Horn, manufactured in 1934, the speaker was opposite the corner, and the built-in horn was a quarter of a 4-foot cinema horn. The bass chamber was also designed differently. Maybe the old drawings were made of this?
Sincerely Tom from Hungary (abonyitamas@freemail.hu)

Dominic Porter 6 years ago · 0 Likes

Which driver did you use in these?
I imagine the originals made have had a succession of Voigt driver evolutions over the production run.
I'd bet if you went for a brand new Lowther you'd probably not want to run the new phase equalizer?

Dominic Porter 6 years ago · 0 Likes

WARNING! Skip 20:00-25:00 in that video!! Unless you like tinnitus?

Dominic Porter 6 years ago · 0 Likes

I think the step is kind of critical to get the HF to point into the room at ear level. The curvature continues past that because the lower frequencies it handles need more expansion. At the very least I suspect that's the logic used in the design. YMMV

Iain 6 years ago · 0 Likes

I wish I had the plans. Pete never made them public. I encouraged them to since they spent so much time on the builds.

Loboear 6 years ago · 0 Likes

Hello , do you have the plans? I would appreciate receiving them . It looks so great.
Best regards from South Germany
Loboear

Anonymous 6 years ago · 0 Likes

Interesting observations from a mind on the matter:
https://www.youtube.com/watch?v=EEh0IPX-q9I

Materials, particularly damping compounds/ techniques and adhesives were the 'realisation of true design' and their importance that boggled the mind on looking closely at John Wrights TDL Reference- his dual confluence transmission line cabinet system.
Special analysis/ attention to original material properties would appear vital for these No.1 Voigts!
The above YouTube link "World's Second Best Speakers!" has good observations regarding these.

micromega 7 years ago · 0 Likes

Dear Iain! I use a Lowther drive for more than twenty years. I would like to build the ultimate Lowther Cabinet The Voigt Domestic Corner Horn. To do that, I would ask for your help, where could the plans be obtained? Thank you in advance for your help. Best Regards: Willem

Iain 8 years ago · 0 Likes

That is really cool. I love that this horn design will live again. If you have pics or info you'd want posted I'd be happy to do it here. If you are going to sell these I'd be happy to promote that as well.

Iain

Unknown 8 years ago · 0 Likes

I have been lucky enough to have borrowed a pair of these horns from a friend for the last few years. With these speakers and thanks to Peter Skinner writing this blog with some great internal photos, I have managed after MANY hours work to create a full set of working CAD drawings in solid works, mm perfect to the original designs. At work I use a large CNC router on a daily basis so at the start of 2018 I can start the construction. I have already created a former for the main reflector, and I am in the process of producing a mould for the concrete reflector. Because of the complex design of these speakers I think using repeatable modern design and construction techniques will make the job a lot easier! It is defiantly true what people who are lucky enough to have heard these speakers say, they truly are something special. Well worth the 250+ hours I have put into getting this far!

saim 8 years ago · 0 Likes

nice post thanks for sharing this blog we at valley trip planner best travel agents based in srinagar kashmir we offer best kashmir tour and travels packages

Iain 8 years ago · 0 Likes

Amazing story there isn't a lot of information out there about Voigt. You were lucky to have been taught by such a legend. Thank you

Gilbert Frederick 8 years ago · 0 Likes

I had Peter Voight as my instructor when I took the electronics course at Ryerson Institute of Technology back in 1951. I was thrilled to hear his loudspeaker at a demonstration he put on at the University of Toronto. He played a violin piece that sounded as if the violin player was actually in the room with us.
During WWII, his speaker was used by the military to listen to broadcasts by the Germans.
He also made a demonstration of his speaker in a large hall in an English city. After playing some classical music, he asked the audience if they knew the wattage of the sound being used to fill the entire theatre - then he held up a small receiving radio tube that was the final output in the amplifier. That showed how efficient the speaker was.

Unknown 9 years ago · 0 Likes

Great post, you have pointed out some excellent points, I as well believe this is a very superb website.
Plastic Flow Meter

Sigitas 9 years ago · 0 Likes

Hello,
Voigt cabinets still available?

pajonec 10 years ago · 0 Likes

Are these plans available yet? would you please contact me pajonec@gmail.com

Anonymous 10 years ago · 0 Likes

My Father had a single corner horn about from 1938 to when we left England in 1949. He was a friend of Voigt and with another friend George Wise used to cut their own records. My Father was a pianist and Georges daughter played the violin. Once settled in Australia he made several horns for a pair of PM7C drivers. It was my intention to build a pair of horns with his drivers. Unfortunately I have to scale down so now cannot do this. I have the drivers if anyone wishes to purchase a pair.
tony.saville@bigpond.com

Unknown 10 years ago · 0 Likes

Hi this is very useful post for everyone. Thank you so much this post. Just about everything looks good displayed. Domestic tour packages

Unknown 11 years ago · 0 Likes

Very well written and informative article, thank you :-)
domestic tour

Unknown 11 years ago · 0 Likes

This comment has been removed by the author.

Iain 11 years ago · 0 Likes

Not 100% sure.
The build is challenging to say the least.
There was a company in Asia selling the reflector. I can't find them now.

Unknown 11 years ago · 0 Likes

And Voigt used a flat canted reflector at the bottom, no?

Unknown 11 years ago · 0 Likes

Hear, hear, Iain. It's nice to interact with all the TRUE BELIEVERS out there. Question for Peter Skinner: Is the diffuser a tractrix curve as well? It's hard to tell from the photos. Voigt could have been preserving the 40Hz cutoff tractrix AREA expansion through the 90 degree bend. And WHY THE STEP? Maybe he was directing and concentrating the beam-like highs in a kind of ear-level annular pattern, while allowing the more diffuse mids to propagate in a more vertically-oriented quarter-spherical distribution. Maybe this was to compensate for a weaker high-frequency response of his early drivers. Seems to me a single curve or even a straight cone segment would work well with modern full-range drivers? I dunno. Aaargh.

Iain 11 years ago · 0 Likes

I think this post gets more comments than anything that has been posted on the site. I hope Peter comes up with the plans. If he sold them on ebay he could at least pay for the drafting.

Unknown 11 years ago · 0 Likes

If I were to do it without drawings, I would try the upper reflector as a simple quarter-cone. Bent mirrors give funny reflections, no?

Unknown 11 years ago · 0 Likes

Been going mental waiting for these drawings. It occurs to me that one might be able to fake the construction, because the bass horn internal shape isn't too critical, quarter-wavelength length should be about 2m, and the upper horn is easily calculated using the tractrix corner expansion that Voigt recommended in 1926. The upper reflector might just be a semi-anti-spherical affair, since it's meant just for reflection, and the step might also be unnecessary, although I don't want to demean the Master, P. G. A. H. Voigt.

Unknown 11 years ago · 0 Likes

When the plans are completed, they will be offered at no charge.
My Blog: jblcharge.blogspot.com

Iain 11 years ago · 0 Likes

Thanks for the update. Can't wait to see another pair. Wish I had the cash for such things. One day perhaps..

Unknown 11 years ago · 0 Likes

Apologies, it's been a year since the blog was posted and we have yet to prepare plans; the reason is that we are going to use the building of a second pair to draw-up accurate plans. But Les has been very busy on other work, and frankly, no-one has yet committed to having a pair built. Given that they would cost about 10,000 euros, that is not surprising. But there's no profit in that figure....just materials, wages and auxillary expenses . I do hope to get plans done, but it isn't guaranteed at the moment.
As for the comment that it wouldn't take 600 hours; that is right. Much of the initial research doesn't need to be replicated. But it will still be a long, difficult build. Indeed, a friend who builds complex speakers for a living, took a look at the Voigt and said that it would take too long to be commercially viable.
As soon as money is available, and les is free, we will be off on the next pair, and plans will be part of that project. We can't really afford to do a second pair on spec.....anyway, thanks for all the kind comments. The Voigt is a great speaker and listening to a pair is an absolute joy.

Unknown 12 years ago · 0 Likes

I wonder whether there are other drawings Lowther Voigt products, such as this - Lowther Hegerman?

http://www.itishifi.com/2010/02/lowther-hegerman-teak-1950-1951-pair.html

Unknown 12 years ago · 0 Likes

Surprising that there are still enthusiasts ! ! !
If you still have this product and beautiful sound , it generally is of great importance . Since it looks exactly Superb! !
I was very impressed with the architect , of course I would like to start this same " build ")))) It is waiting for the plans , I would be happy to help make free drawings, but I live in Ukraine in Odessa))) By the way, at least as something to offset the costs authors can publish a book about the history of drawings and photos . would love to have bought this . Only one on the internet (to start as soon as possible to build)))) , and the other would wait by mail))))

With great respect , Serge

Unknown 12 years ago · 0 Likes

This comment has been removed by the author.

Unknown 12 years ago · 0 Likes

This comment has been removed by the author.

Unknown 12 years ago · 0 Likes

mindless drivel...no reason these should take 600 man hours after the first batch are made

Justin Benn 12 years ago · 0 Likes

Are these plans available yet?

Iain 12 years ago · 0 Likes

I have not heard if plans are available yet. When they are available I'm sure Peter will contact us for another post.

Unknown 12 years ago · 0 Likes

What a great build do you know if the plans are available yet, I am looking for a long term project with my Father, who can do sculpture and quality woodwork, having build some simpler projects together.

Thanks
Simon, I

Bromo33333 12 years ago · 0 Likes

Wonderful!

Unknown 12 years ago · 0 Likes

I'm interested in ordering a pair, please contact me off line with pricing and timing.
david

Iain 12 years ago · 0 Likes

They have done the hard part. I like the kick starter idea or even a small charge for the plans.

Owen C McPeake 12 years ago · 0 Likes

I want these plans!

Owen C McPeake 12 years ago · 0 Likes

This comment has been removed by the author.

Hornloco 12 years ago · 0 Likes

René Monday, July 29th 2013

Marvellous and courageous undertaking, relatively swiftly and successfully driven to completion : bravo !

In 1954, I was completing my last year of electronic engineering study at the polytechnic school in Lausanne, Switzerland. A "rich" student-friend of mine had purchased a Voigt Domestic Corner Horn in London and imported it into Switzerland. I fell at once and completely under the spell of this speaker which sounded so natural, on classical music particularly.

At the time I was already a convinced audiophile but it demonstrated readily to
me the magic of the horn as an acoustic transducer. The Voigt Horn did also
show me the advantage offered by an even distribution of the sound energy
in the horizontal plane : no or little sweet spot effect. This was not so important in the « mono-era »
but the situation changed drastically with the coming of stereo !

Reply

Unknown 12 years ago · 0 Likes

Amazing breakdown! I can't quite visualize the sound path but I'd love to hear it.

Anonymous 12 years ago · 0 Likes

So do we know roughly how much 600+ man hours costs? Also was that per speaker or for the pair?

One day I'll be rich...